PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 7 Submission

Name of the Student	Pranav Hemanth
SRN	PES1UG23CS433
Section	G
Department	CSE
Campus	RR

Topic: Lab 7: Email Forensics

UE23CS342BA6

Objective:

The objective of this lab is to analyze email files to identify legitimate and malicious emails. Students will perform email header analysis to detect phishing and CEO Fraud (BEC) attacks using standard forensic tools.

Extract and interpret key email header fields (From, Return-Path, Received, X-Originating-IP). Perform SPF, DKIM, and DMARC validation using MXToolbox. Classify each evidence email as Legitimate, Phishing, or Business Email Compromise (BEC). Document findings in a structured investigative format.

Required software:

Tool	Purpose	Access
Mozilla Thunderbird	Open and inspect .eml evidence files	www.thunderbird.net (Free)
MXToolbox Email Analyzer	Parse headers — SPF, DKIM, DMARC, IP	mxtoolbox.com/EmailHeaders.aspx
Whois / IP Lookup	Trace and geolocate the originating IP	whois.domaintools.com

Evidence Files:

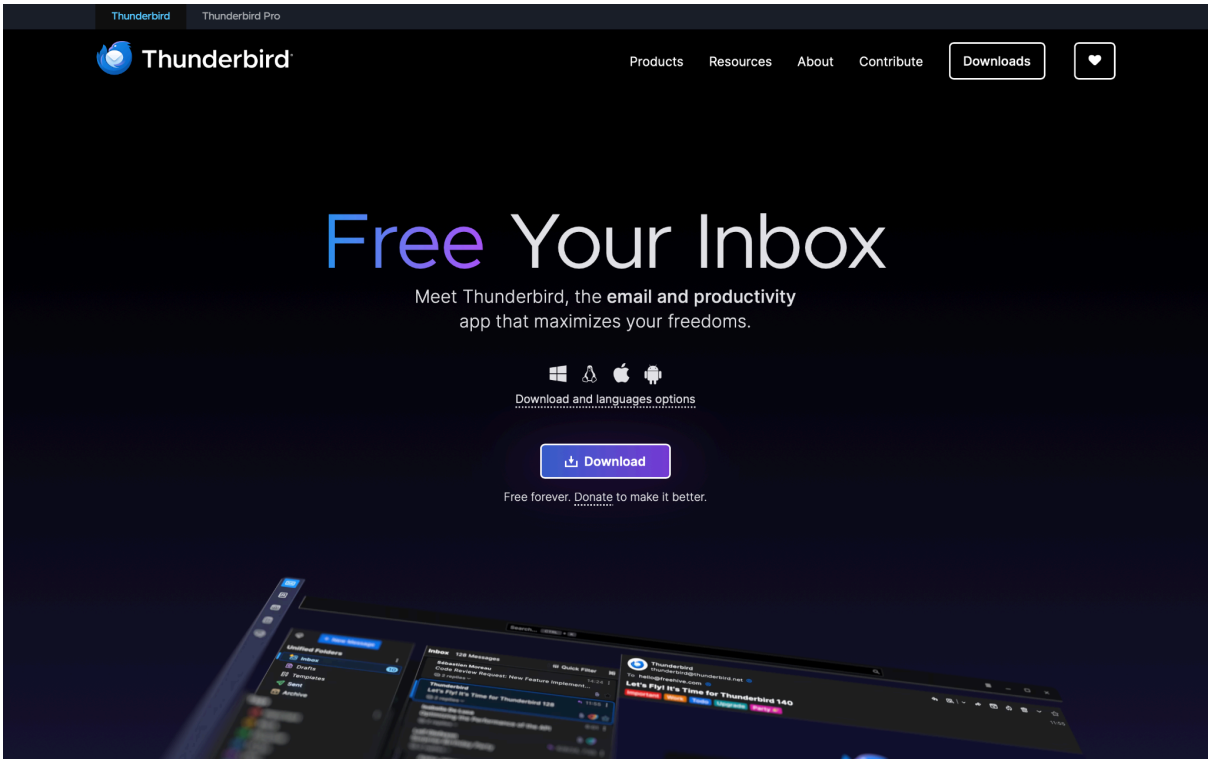
Three email evidence files are provided. Do not modify the originals. Open each file read-only.

ID	Email Type	Threat Category
EV01	Legitimate Business Email	Baseline / Control Sample
EV02	Phishing Email	Credential Harvesting Attack
EV03	CEO Fraud Email	Business Email Compromise (BEC)

Installation and Labsetup:

Step1: Do the following for Mozilla Thunderbird

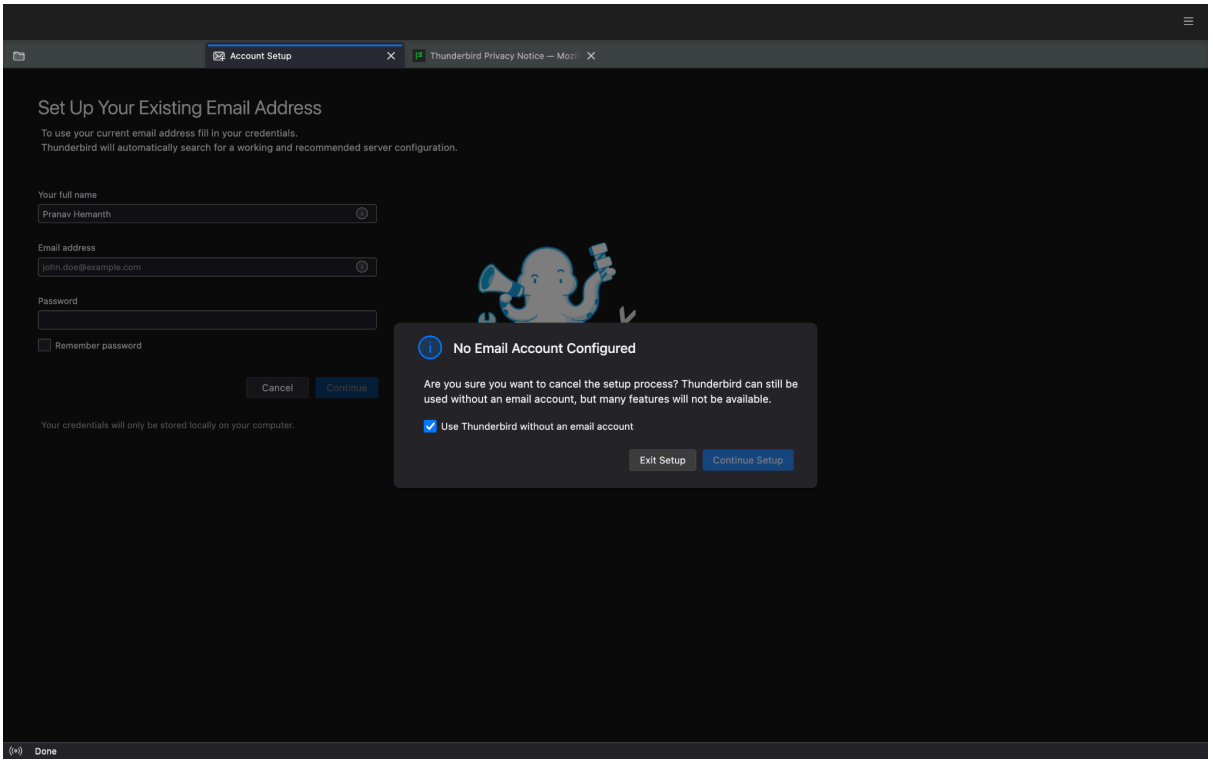
1) Download Thunderbird from www.thunderbird.net.



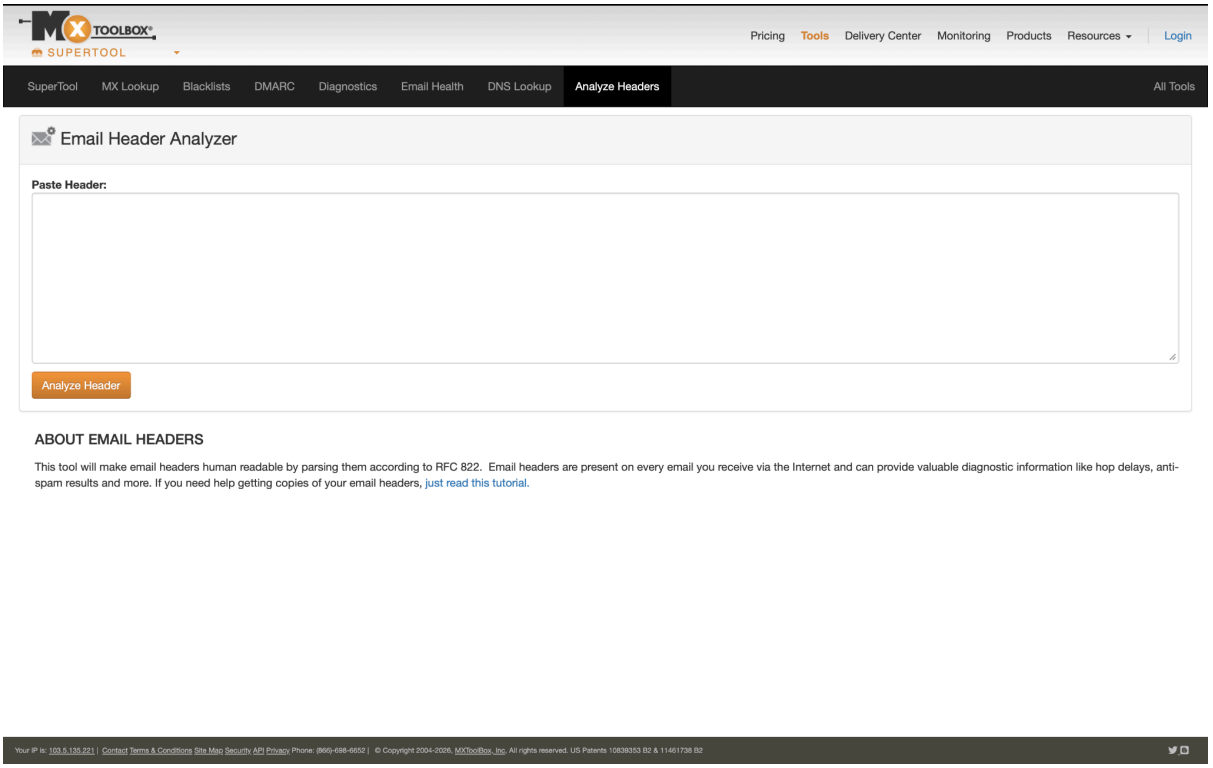
2) Run the installer for Thunderbird



3) On first launch click 'Skip this and use existing email' to bypass account setup.



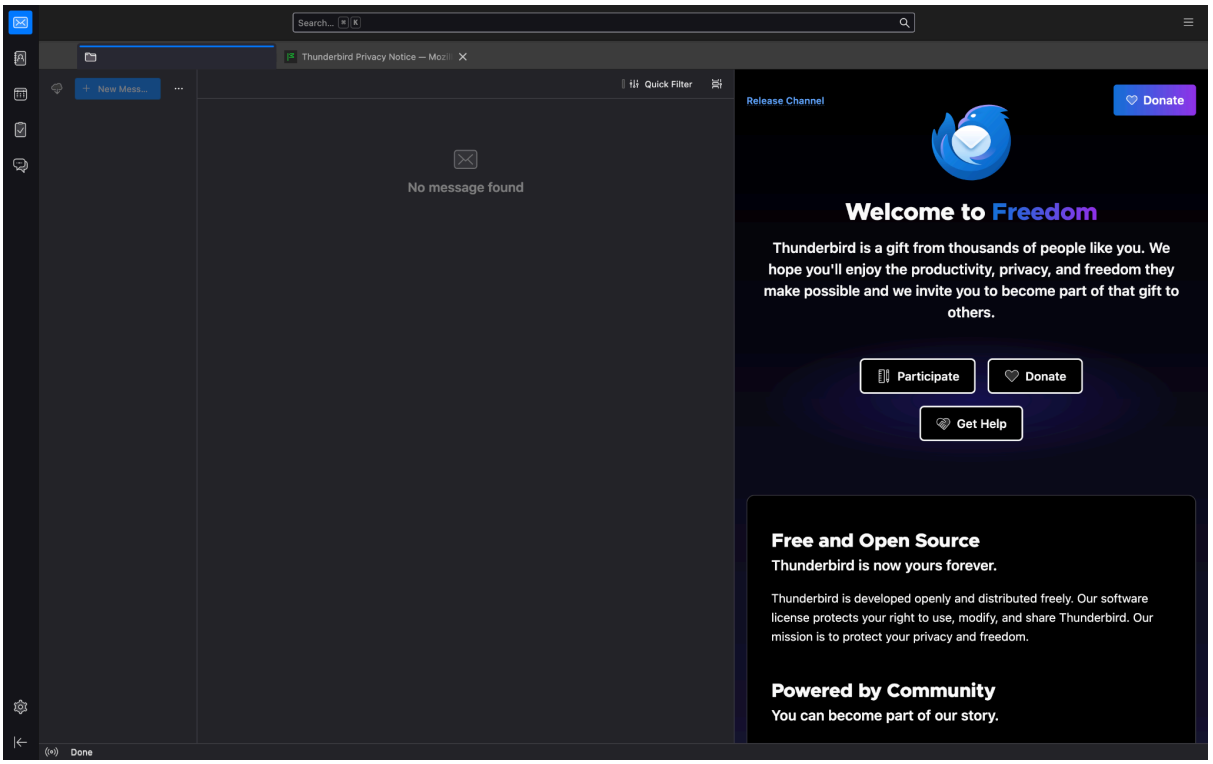
4) Open a browser and confirm access to https://mxtoolbox.com/EmailHeaders.aspx.



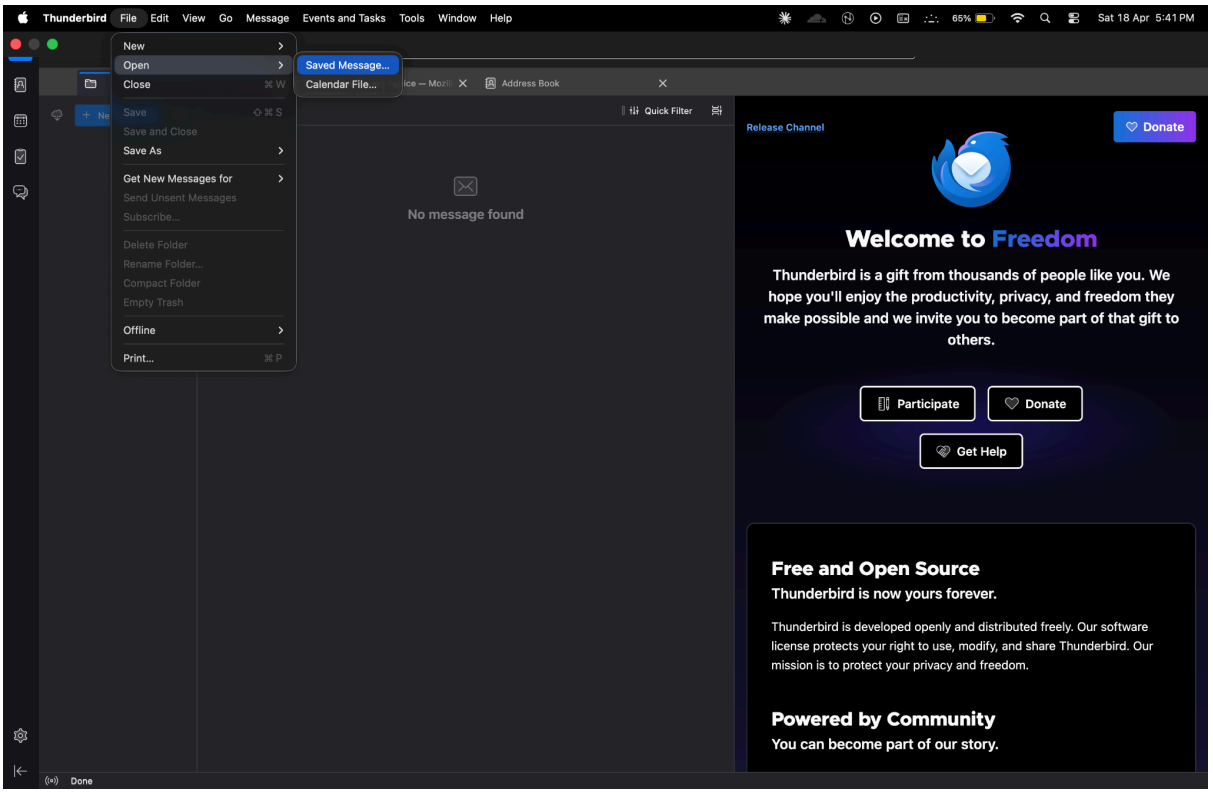
Procedure to perform the lab:

Step1: Open Evidence Files in Thunderbird

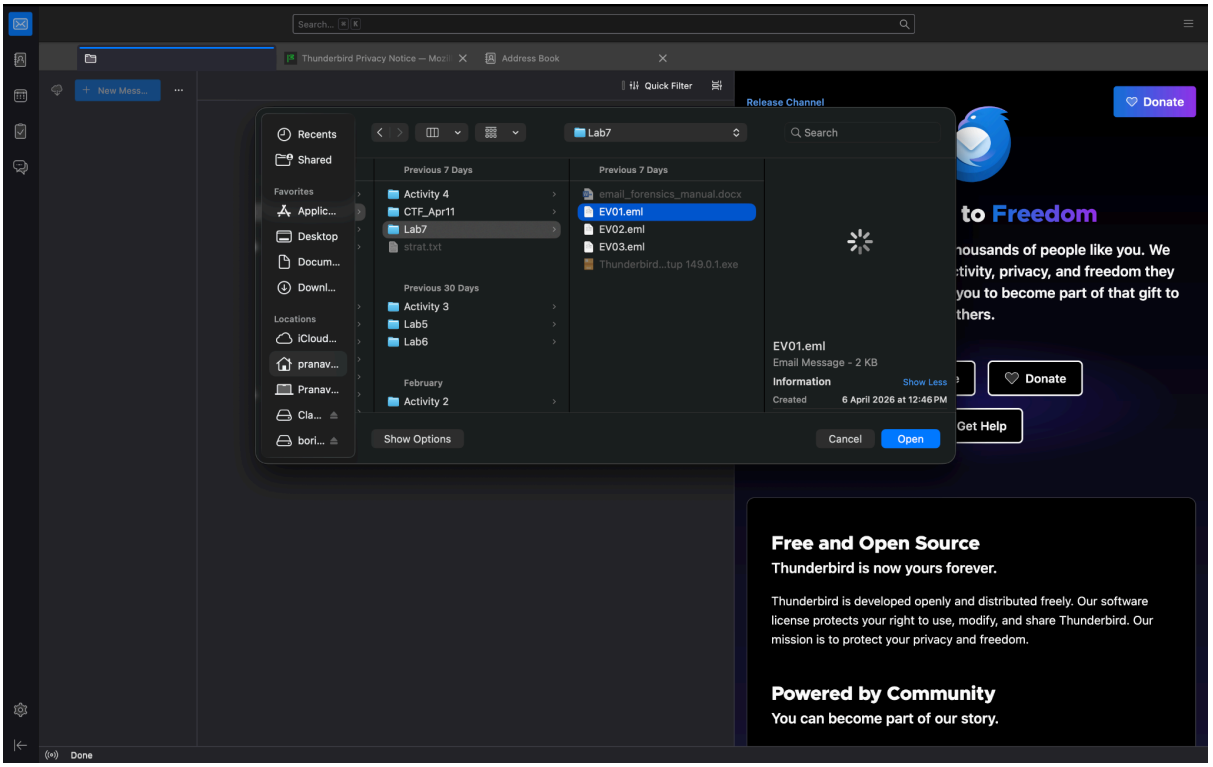
5) Launch Mozilla Thunderbird

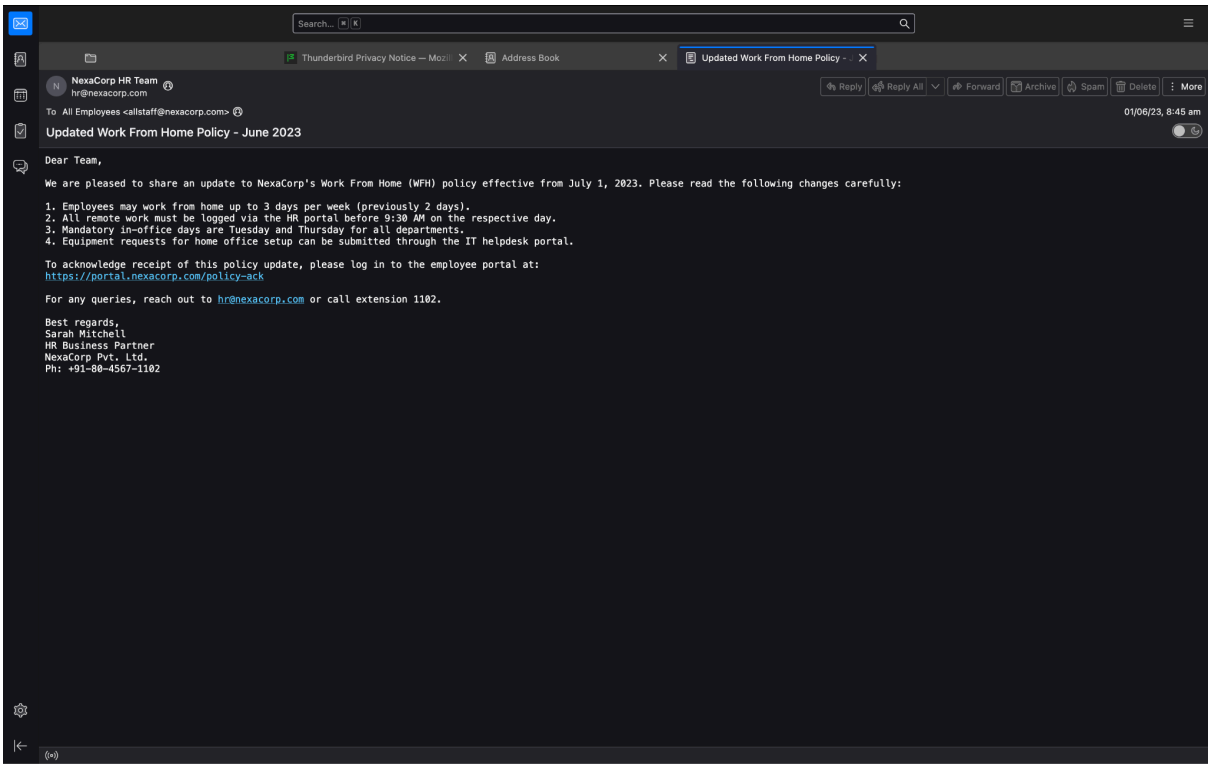


6) Go to File > Open > Saved Message



7) Select EV01.eml from the evidence folder and click Open.

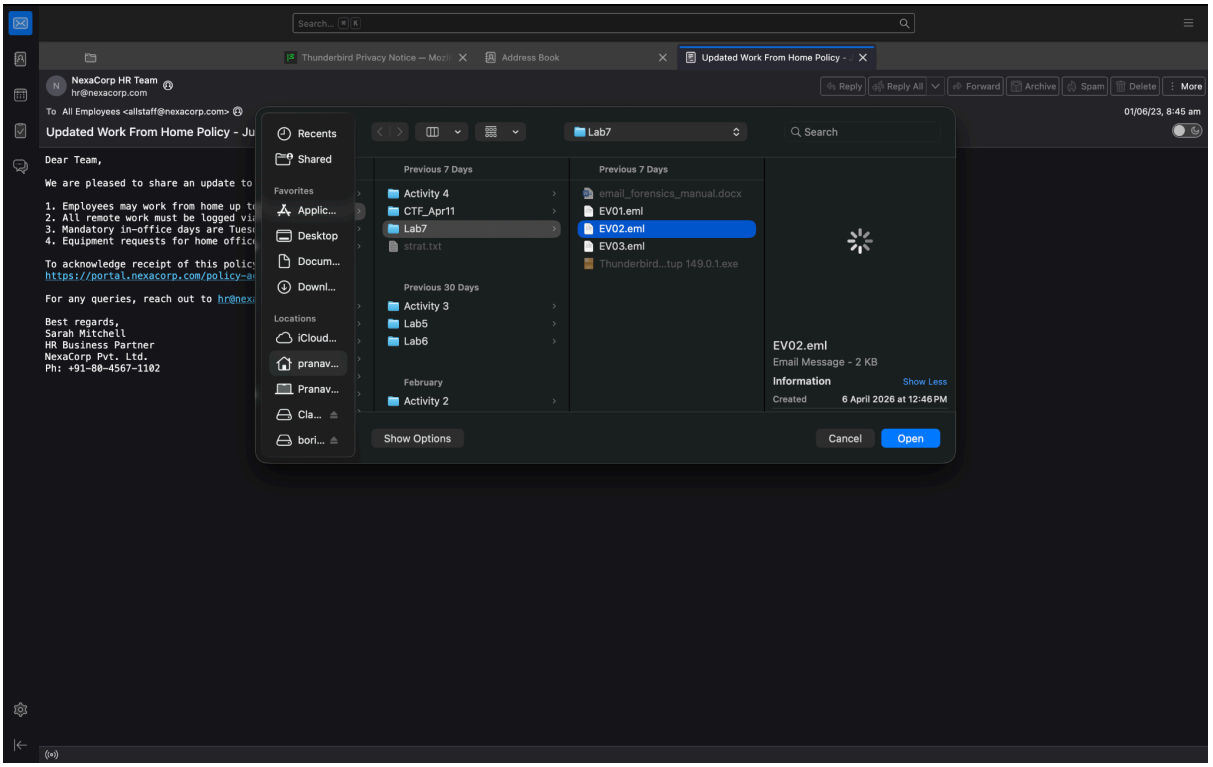


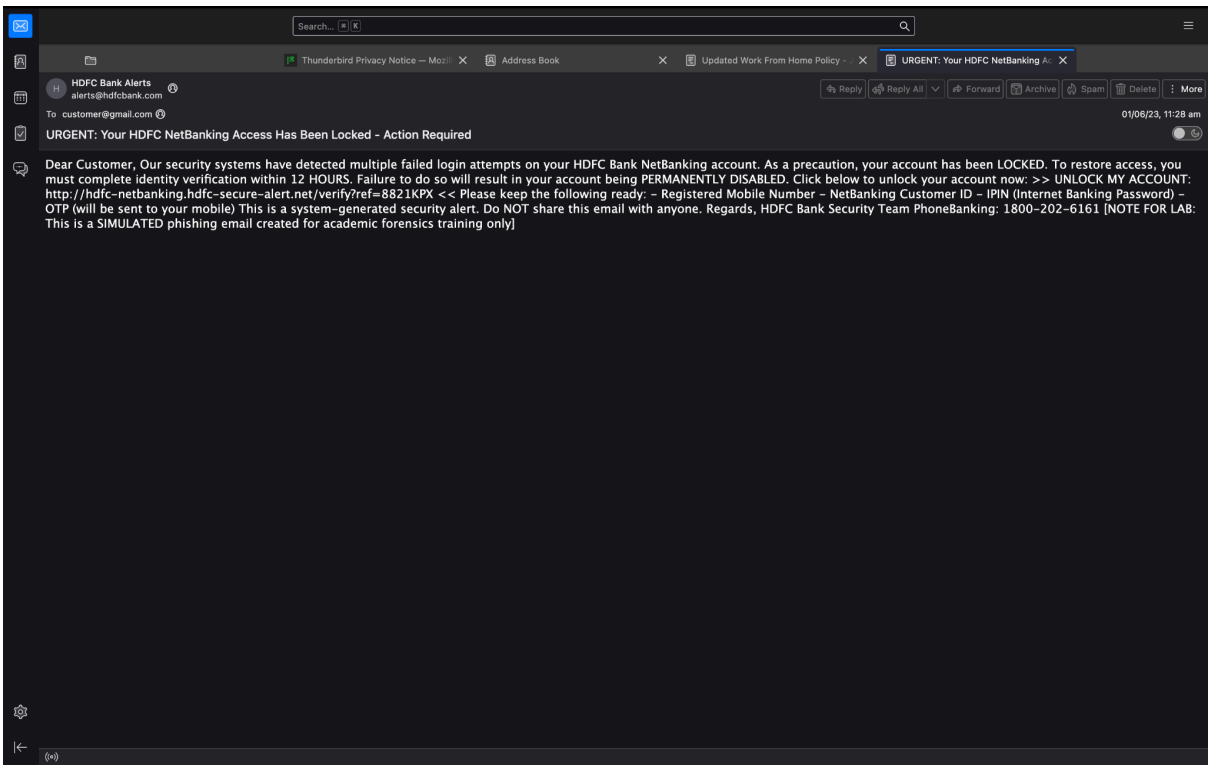


8) Note the displayed From, To, Subject, and Date fields.

From: NexaCorp HR Team <hr@nexacorp.com>
To: All Employees <allstaff@nexacorp.com>
Subject: Updated Work From Home Policy - June 2023
Date: 6/1/2023, 08:45

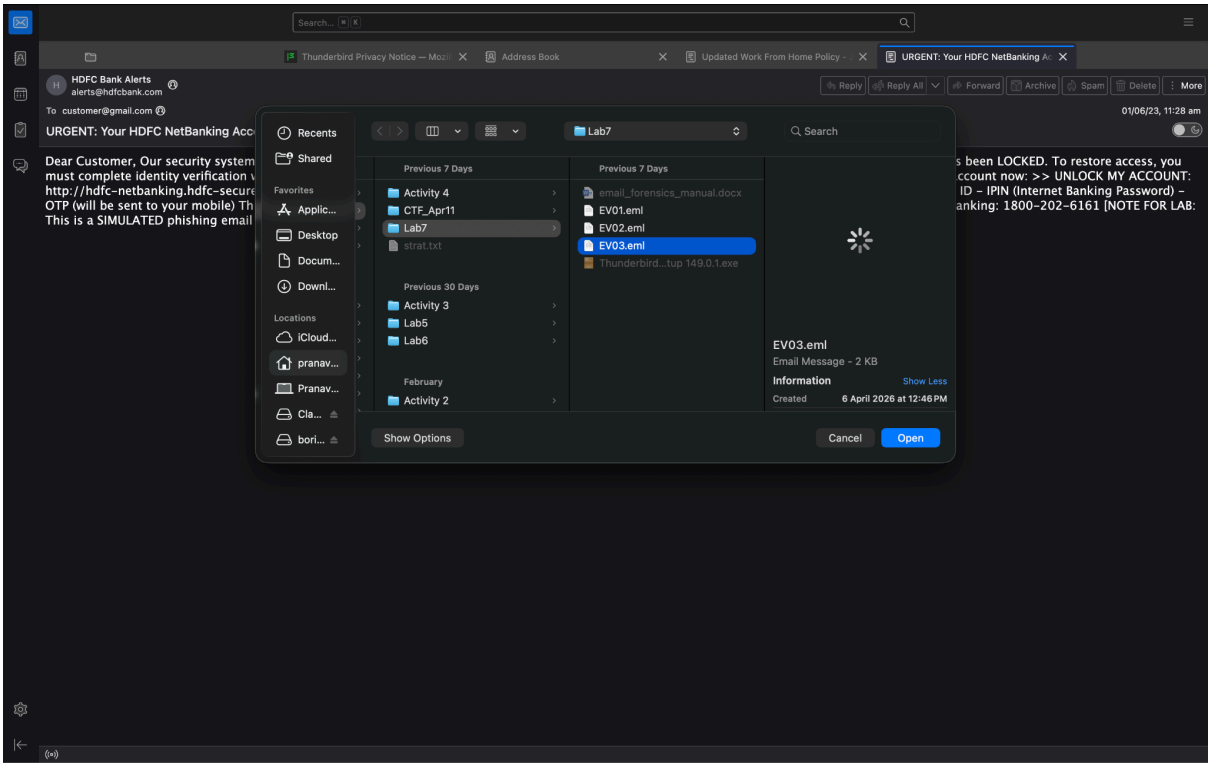
9) Repeat for EV02.eml.

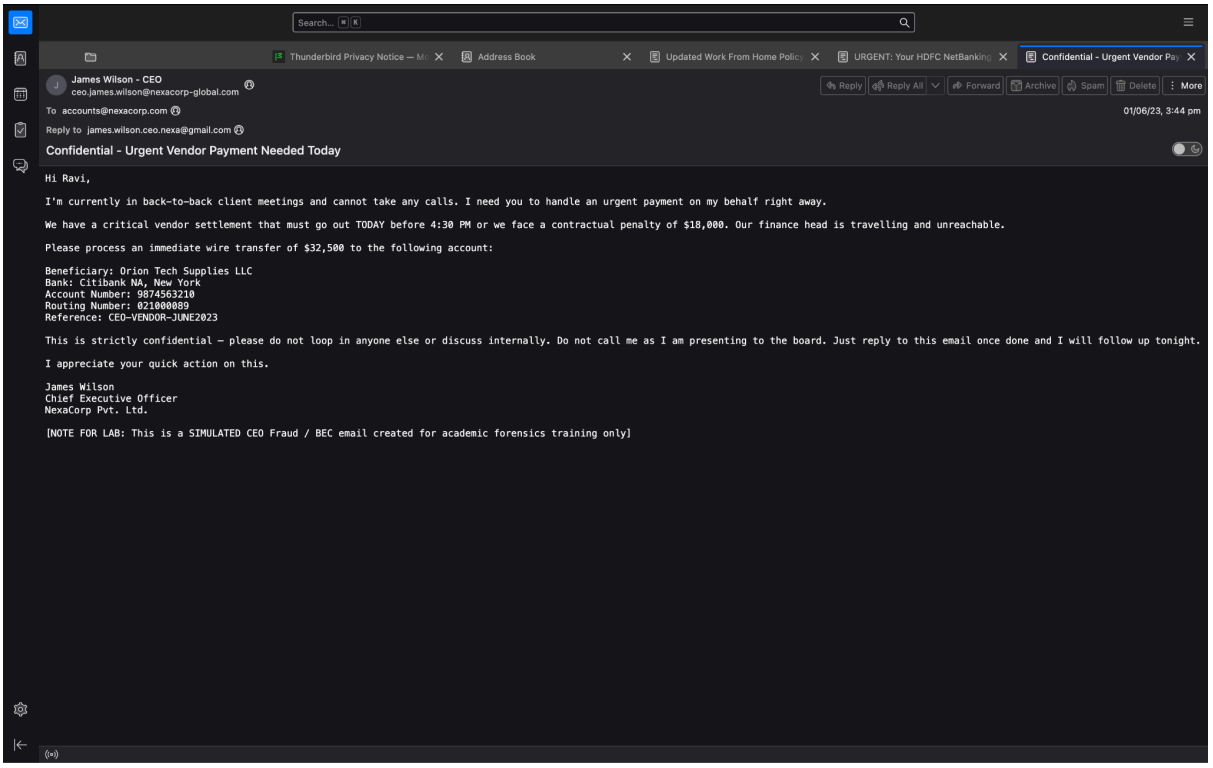




From: HDFC Bank Alerts <alerts@hdfcbank.com>
To: customer@gmail.com
Subject: URGENT: Your HDFC NetBanking Access Has Been Locked - Action Required
Date: 6/1/2023, 11:28

10)Repeat for EV03.eml.



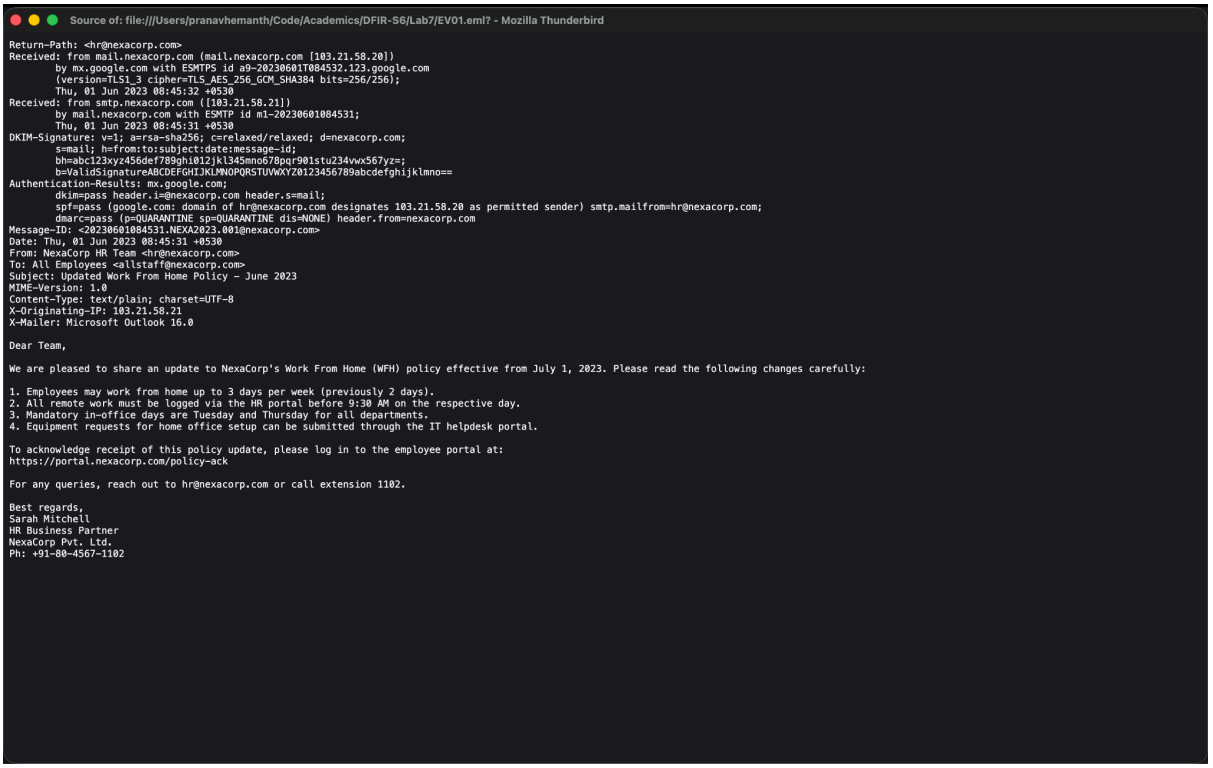


From: James Wilson - CEO <ceo.james.wilson@nexacorp-global.com>
To: accounts@nexacorp.com
Subject: Confidential - Urgent Vendor Payment Needed Today
Date: 6/1/2023, 15:44

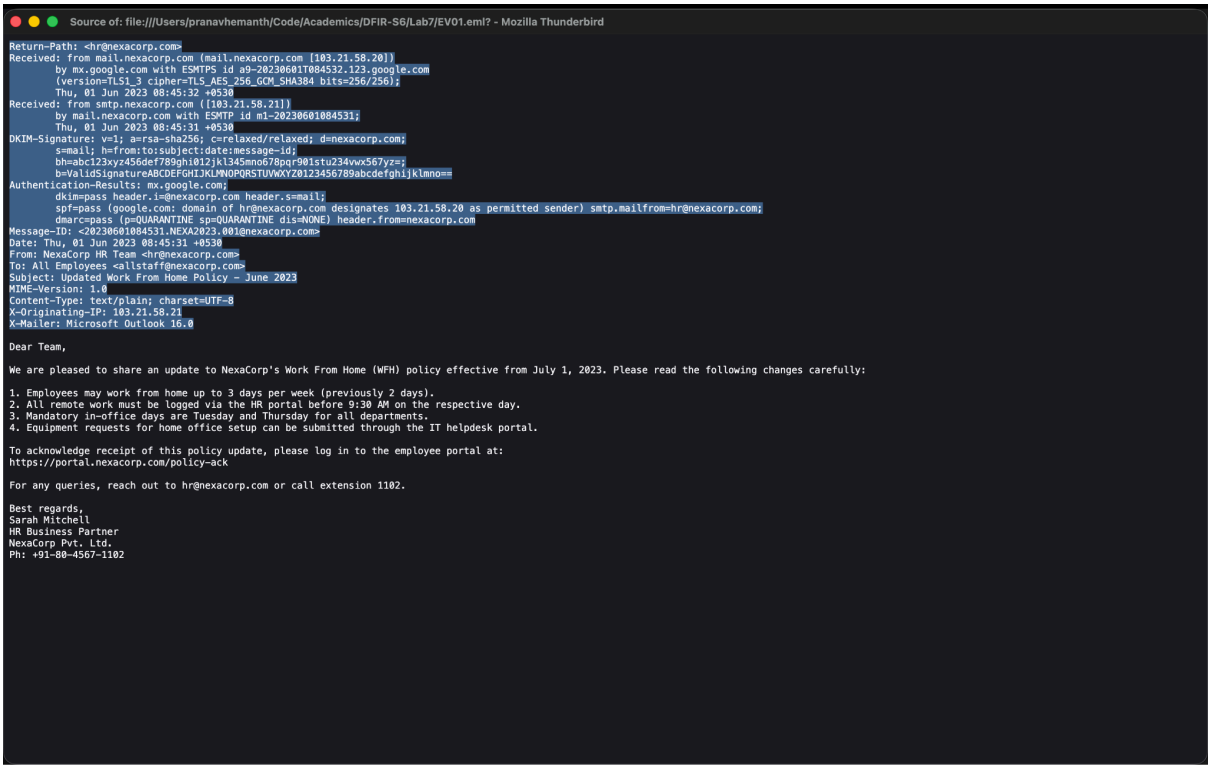
Step2: Extract Raw Email Headers

11) With an evidence file open, press Cmd + U to open the raw message source.

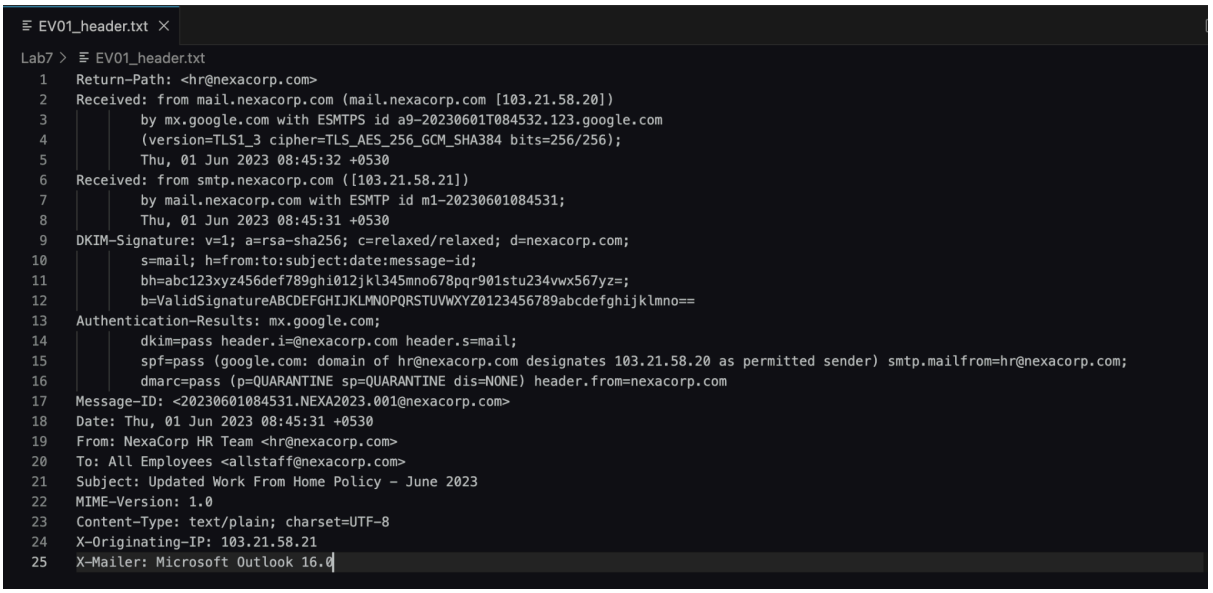
12) Extract the headers from EV01.eml



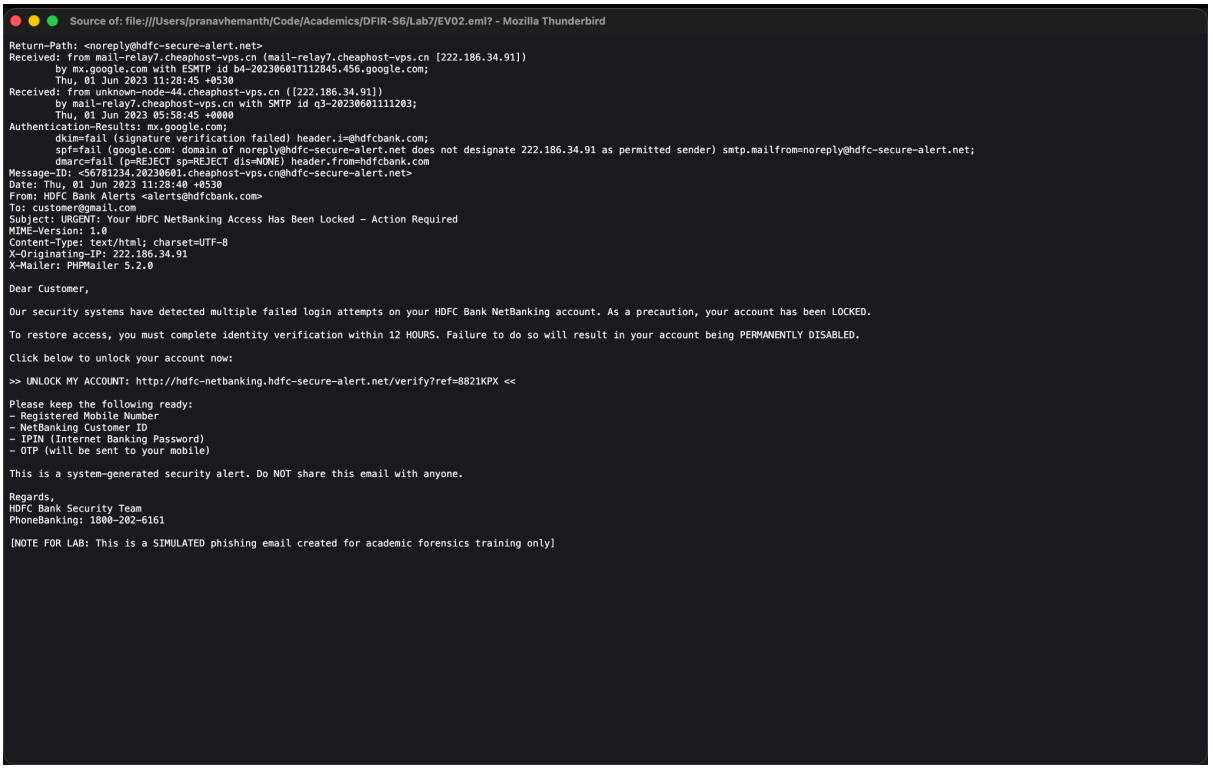
13) Select all text from the first header field down to the blank line before the email body.

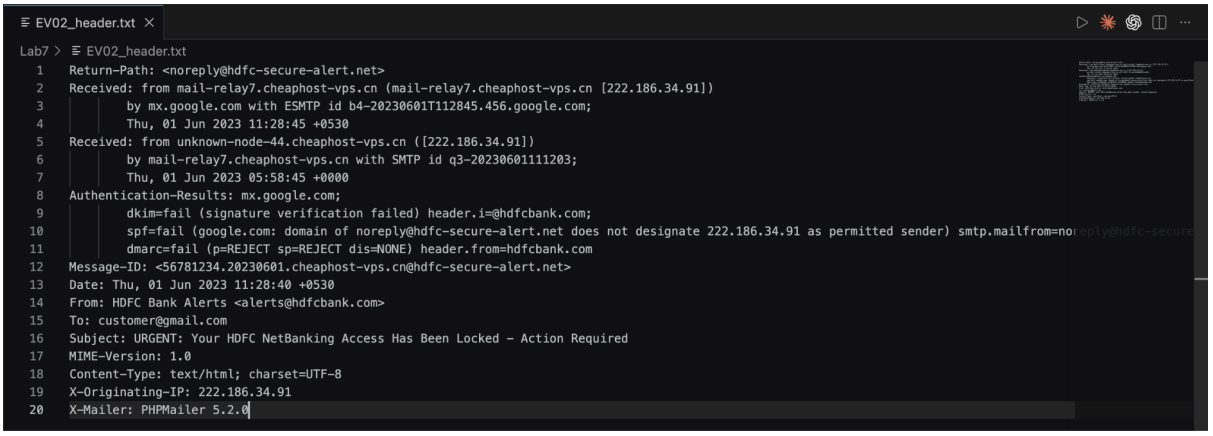


14)Copy the header text and paste it into Notepad. Save as EV01_header.txt (repeat for each file).

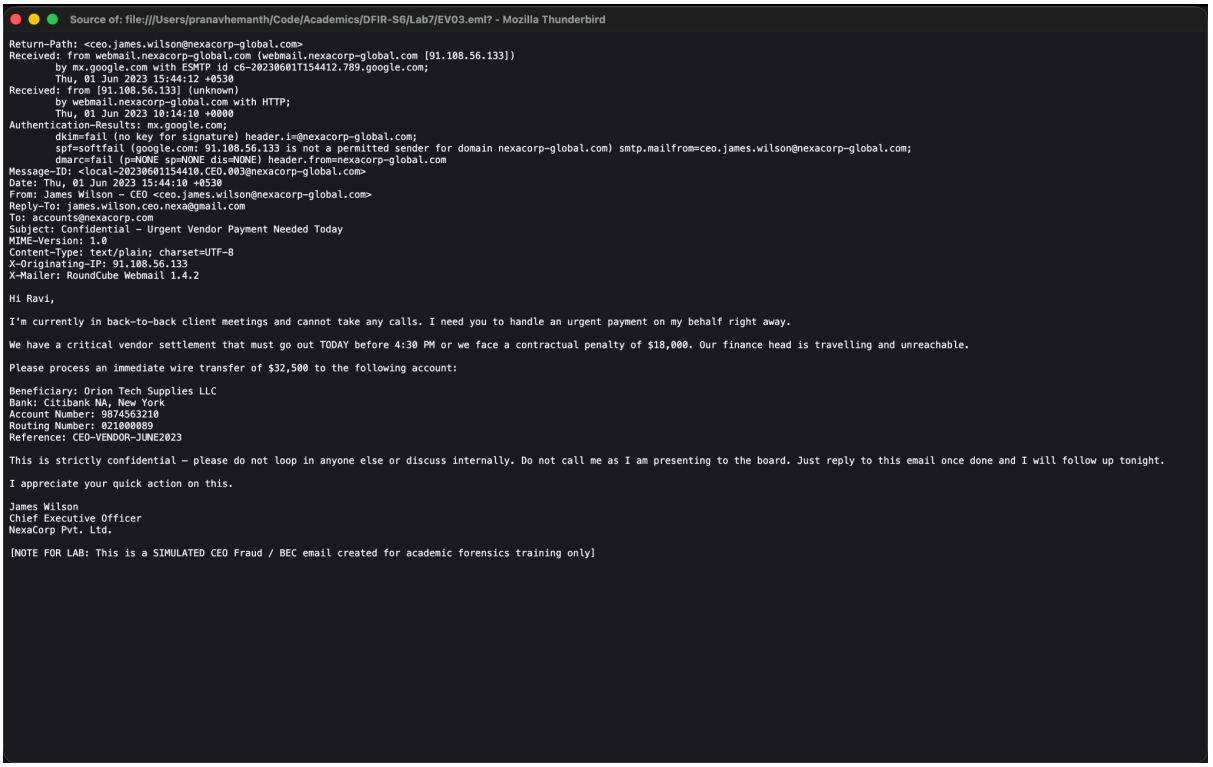


15)Extract the headers from EV02.eml





16)Extract the headers from EV03.eml



Source of: file:///Users/pranavhemanth/Code/Academics/DFIR-S6/Lab7/EV03.eml? - Mozilla Thunderbird

```
Return-Path: <ceo.james.wilson@nexacorp-global.com>
Received: from webmail.nexacorp-global.com (webmail.nexacorp-global.com [91.108.56.133])
        by mx.google.com with ESMTP id c6-20230601T154412.789.google.com;
        Thu, 01 Jun 2023 15:44:12 +0530
Received: from [91.108.56.133] (unknown)
        by webmail.nexacorp-global.com with HTTP;
        Thu, 01 Jun 2023 10:14:10 +0000
Authentication-Results: mx.google.com;
        dkim=fail (no key for signature) header.i=@nexacorp-global.com;
        spf=softfail (google.com: 91.108.56.133 is not a permitted sender for domain nexacorp-global.com) smtp.mailfrom=ceo.james.wilson@nexacorp-global.com;
        dmarc=fail (p=NONE sp=NONE dis=NONE) header.from=nexacorp-global.com
Message-ID: <local-20230601154410.CEO.003@nexacorp-global.com>
Date: Thu, 01 Jun 2023 15:44:10 +0530
From: James Wilson - CEO <ceo.james.wilson@nexacorp-global.com>
Reply-To: james.wilson.ceo.nexa@gmail.com
To: accounts@nexacorp.com
Subject: Confidential - Urgent Vendor Payment Needed Today
MIME-Version: 1.0
Content-Type: text/plain; charset=UTF-8
X-Originating-IP: 91.108.56.133
X-Mailer: RoundCube Webmail 1.4.2

Hi Ravi,

I'm currently in back-to-back client meetings and cannot take any calls. I need you to handle an urgent payment on my behalf right away.

We have a critical vendor settlement that must go out TODAY before 4:30 PM or we face a contractual penalty of $18,000. Our finance head is travelling and unreachable.

Please process an immediate wire transfer of $32,500 to the following account:

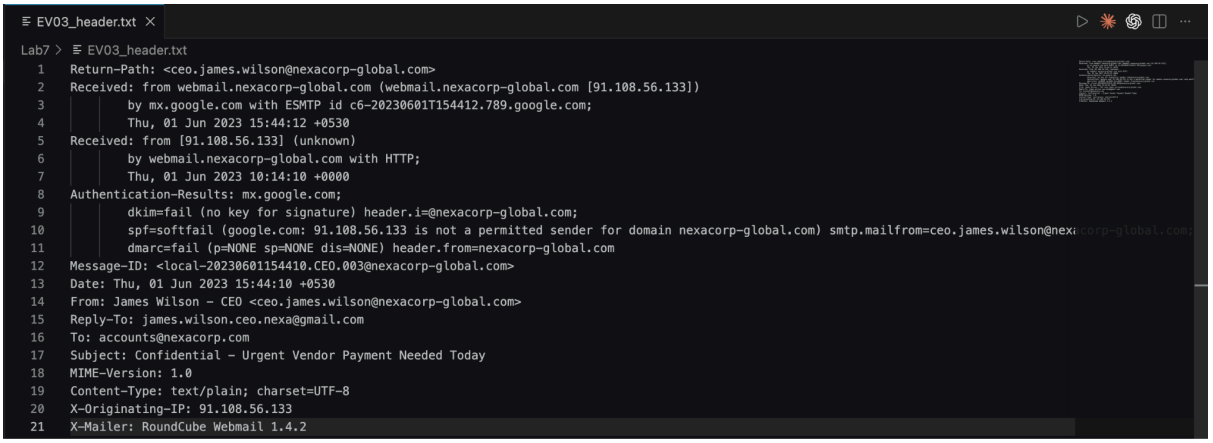
Beneficiary: Orion Tech Supplies LLC
Bank: Citibank NA, New York
Account Number: 9874563210
Routing Number: 021000089
Reference: CEO-VENDOR-JUNE2023

This is strictly confidential - please do not loop in anyone else or discuss internally. Do not call me as I am presenting to the board. Just reply to this email once done and I will follow up tonight.

I appreciate your quick action on this.

James Wilson
Chief Executive Officer
NexaCorp Pvt. Ltd.

[NOTE FOR LAB: This is a SIMULATED CEO Fraud / BEC email created for academic forensics training only]
```



EV03_header.txt

Lab7 > EV03_header.txt

```
1   Return-Path: <ceo.james.wilson@nexacorp-global.com>
2   Received: from webmail.nexacorp-global.com (webmail.nexacorp-global.com [91.108.56.133])
3           by mx.google.com with ESMTP id c6-20230601T154412.789.google.com;
4           Thu, 01 Jun 2023 15:44:12 +0530
5   Received: from [91.108.56.133] (unknown)
6           by webmail.nexacorp-global.com with HTTP;
7           Thu, 01 Jun 2023 10:14:10 +0000
8   Authentication-Results: mx.google.com;
9           dkim=fail (no key for signature) header.i=@nexacorp-global.com;
10          spf=softfail (google.com: 91.108.56.133 is not a permitted sender for domain nexacorp-global.com) smtp.mailfrom=ceo.james.wilson@nexacorp-global.com;
11          dmarc=fail (p=NONE sp=NONE dis=NONE) header.from=nexacorp-global.com
12  Message-ID: <local-20230601154410.CEO.003@nexacorp-global.com>
13  Date: Thu, 01 Jun 2023 15:44:10 +0530
14  From: James Wilson - CEO <ceo.james.wilson@nexacorp-global.com>
15  Reply-To: james.wilson.ceo.nexa@gmail.com
16  To: accounts@nexacorp.com
17  Subject: Confidential - Urgent Vendor Payment Needed Today
18  MIME-Version: 1.0
19  Content-Type: text/plain; charset=UTF-8
20  X-Originating-IP: 91.108.56.133
21  X-Mailer: RoundCube Webmail 1.4.2
```

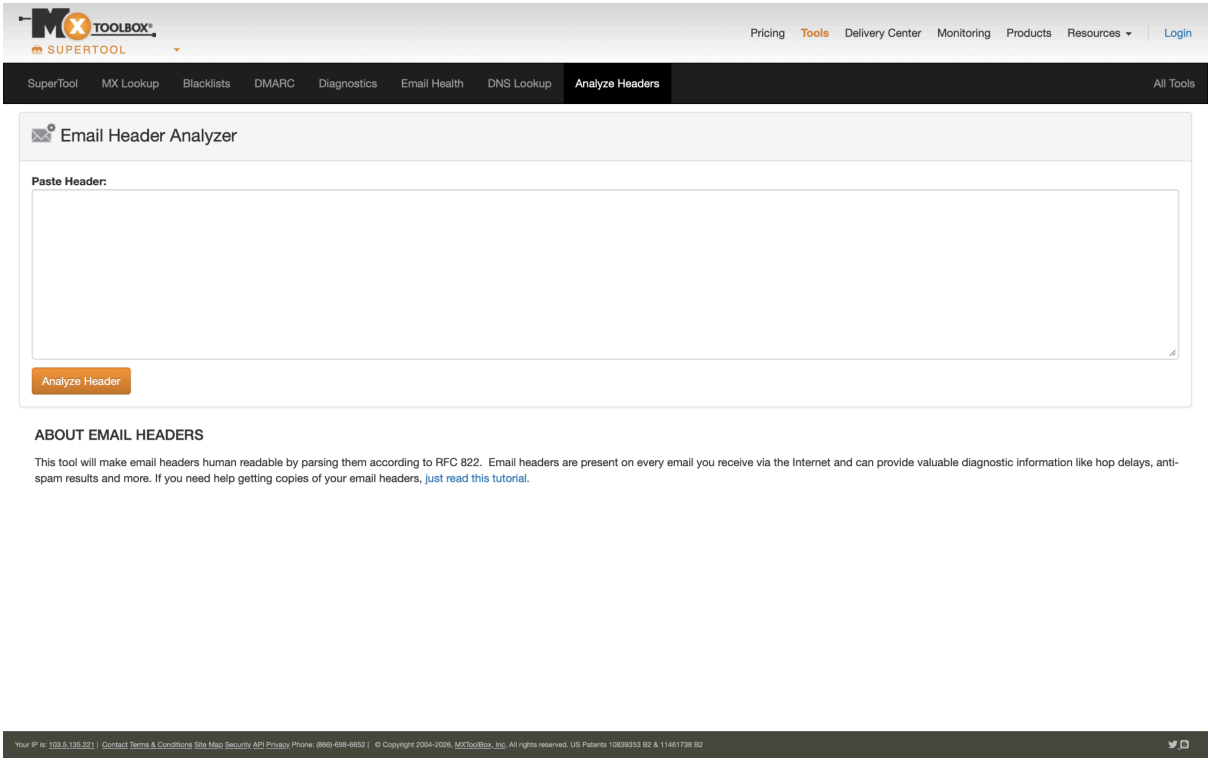
17)Verify the files:

```
pranavhemanth@Pranavs-MacBook-Pro-M3 DFIR-S6 %cd Lab7
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab7 %tree
.
├── email_forensics_manual.docx
├── EV01_header.txt
├── EV01.eml
├── EV02_header.txt
├── EV02.eml
├── EV03_header.txt
├── EV03.eml
└── Thunderbird Setup 149.0.1.exe

1 directory, 8 files
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab7 %
```

Step3: Analyze Headers Using MXToolbox

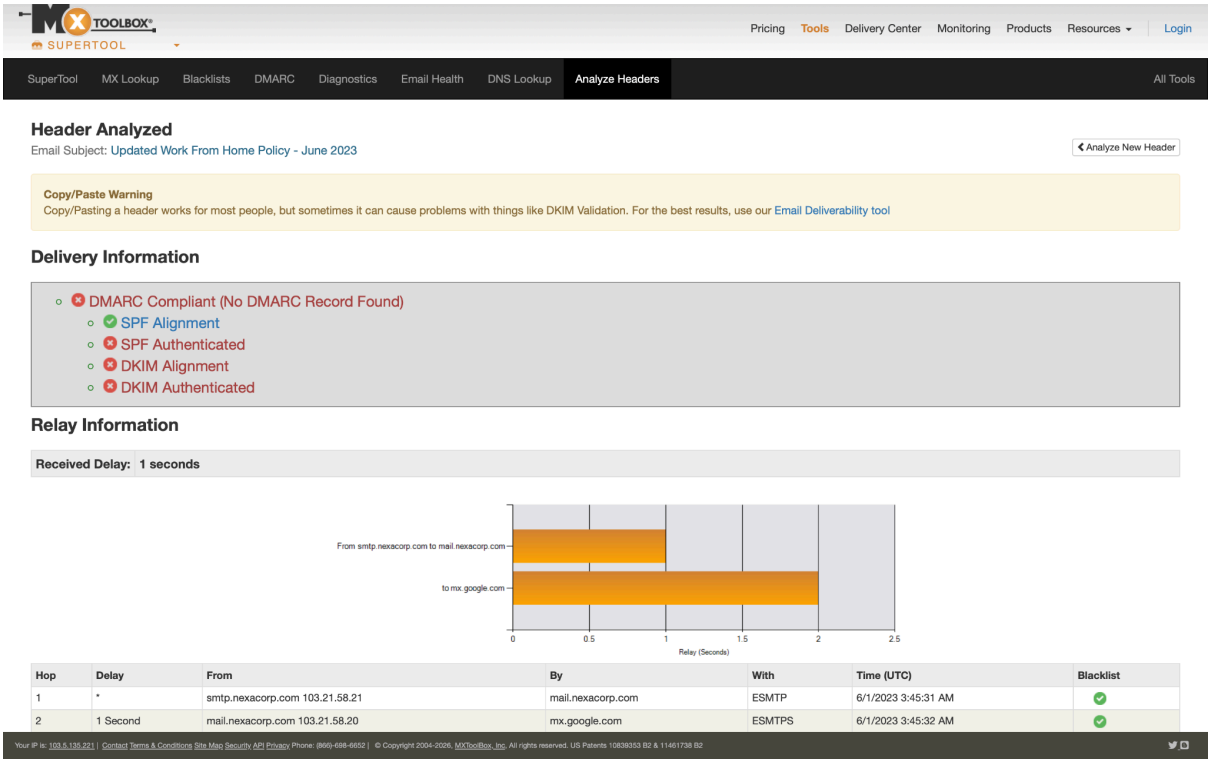
18)Go to https://mxtoolbox.com/EmailHeaders.aspx.



19)Do the below for all 3 files:

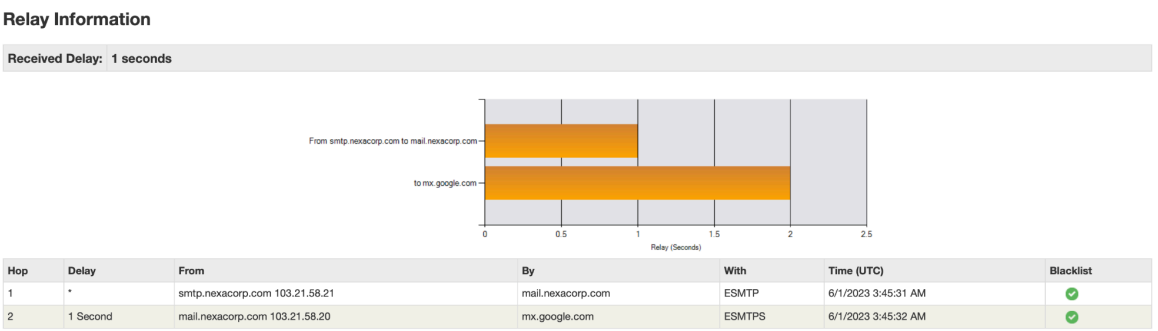
EV01.eml

20)Paste the copied header into the input field and click Analyze Header.



21)Record the following from the results for each evidence file:

a) Originating IP Address and its geographic location.



- smtp.nexacorp.com 103.21.58.21
- mail.nexacorp.com 103.21.58.20

Whois IP 103.21.58.21

Updated 54 seconds ago

```
% [whois.apnic.net]
% Whois data copyright terms    http://www.apnic.net/db/dbcopyright.html

% Information related to '103.21.58.0 - 103.21.58.255'

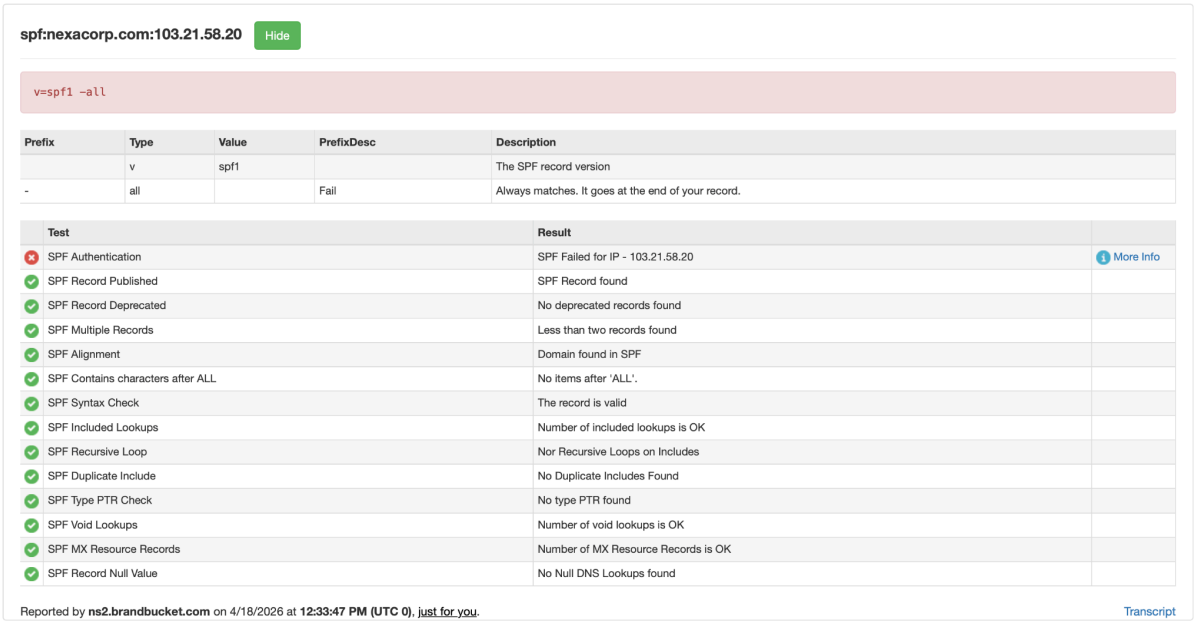
% Abuse contact for '103.21.58.0 - 103.21.58.255' is 'abuse@publicdomainregistry.com'

inetnum:        103.21.58.0 - 103.21.58.255
netname:        PDRSOLUTIONSFZC-AP
descr:          P.D.R Solutions FZC
country:        IN
geoloc:         19.1140343 72.8921789
admin-c:        PSFA1-AP
tech-c:         PSFA1-AP
abuse-c:        AI346-AP
status:         ALLOCATED NON-PORTABLE
mnt-by:         MAINT-IN-AP
mnt-irt:        IRT-IN-GPX
last-modified:  2021-01-06T13:12:21Z
source:         APNIC

irt:            IRT-IN-GPX
address:        GPX India. Unit A-001 Boomerang Chandivali Farm Road Andheri East, Mumbai 400072,
e-mail:         ipadmin@publicdomainregistry.com
abuse-mailbox:  abuse@publicdomainregistry.com
admin-c:        PSFA1-AP
tech-c:         PSFA1-AP
auth:           # Filtered
remarks:        abuse@publicdomainregistry.com was validated on 2025-12-20
```

- Geographic location: Mumbai, India

b) SPF Result — Pass / Fail (fail = unauthorized sender).



spf:nexacorp.com:103.21.58.20 Hide

`v=spf1 -all`

Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
-	all		Fail	Always matches. It goes at the end of your record.

	Test	Result	
✗	SPF Authentication	SPF Failed for IP - 103.21.58.20	More Info
✓	SPF Record Published	SPF Record found	
✓	SPF Record Deprecated	No deprecated records found	
✓	SPF Multiple Records	Less than two records found	
✓	SPF Alignment	Domain found in SPF	
✓	SPF Contains characters after ALL	No items after 'ALL'.	
✓	SPF Syntax Check	The record is valid	
✓	SPF Included Lookups	Number of included lookups is OK	
✓	SPF Recursive Loop	Nor Recursive Loops on Includes	
✓	SPF Duplicate Include	No Duplicate Includes Found	
✓	SPF Type PTR Check	No type PTR found	
✓	SPF Void Lookups	Number of void lookups is OK	
✓	SPF MX Resource Records	Number of MX Resource Records is OK	
✓	SPF Record Null Value	No Null DNS Lookups found	

Reported by **ns2.brandbucket.com** on 4/18/2026 at **12:33:47 PM (UTC 0)**, just for you. Transcript

- Fail - the sender is unauthorised

c) DKIM Result — Pass / Fail (fail = message may be altered).

Dkim Signature Error:
DKIM-Signature Domain dkim:nexacorp.com:mail is invalid – more info

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. – more info

Headers Found

Header Name	Header Value
Return-Path	<hr@nexacorp.com>
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=nexacorp.com; s=mail; h=from:to:subject:date:message-id; bh=abc123xyz456def789ghi012jkl345mno678pqr901stu234vwx567yz=; b=ValidSignatureABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789abcdefghijklmno==
Authentication-Results	mx.google.com; dkim=pass header.i=@nexacorp.com header.s=mail; spf=pass (google.com: domain of hr@nexacorp.com designates 103.21.58.20 as permitted sender) smtp.mailfrom=hr@nexacorp.com; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=nexacorp.com
Message-ID	<20230601084531.NEXA2023.001@nexacorp.com>
Date	Thu, 01 Jun 2023 08:45:31 +0530
From	NexaCorp HR Team <hr@nexacorp.com>
To	All Employees <allstaff@nexacorp.com>
Subject	Updated Work From Home Policy - June 2023
MIME-Version	1.0
Content-Type	text/plain; charset=UTF-8
X-Originating-IP	103.21.58.21
X-Mailer	Microsoft Outlook 16.0

- Fail - the message may be altered

d) DMARC Result — Pass / Fail and the domain policy.

dmarc:nexacorp.com Hide | Solve Email Delivery Problems

Test	Result	
DMARC Record Published	No DMARC Record found	More Info

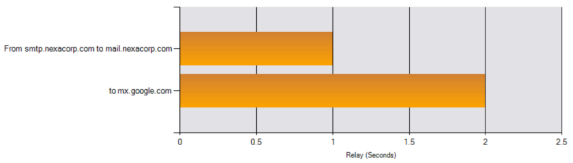
Reported by **ns2.brandbucket.com** on 4/18/2026 at **12:33:47 PM (UTC 0)**, just for you. Transcript

- Fail

e) Number of relay hops in the Received: chain.

Relay Information

Received Delay: 1 seconds



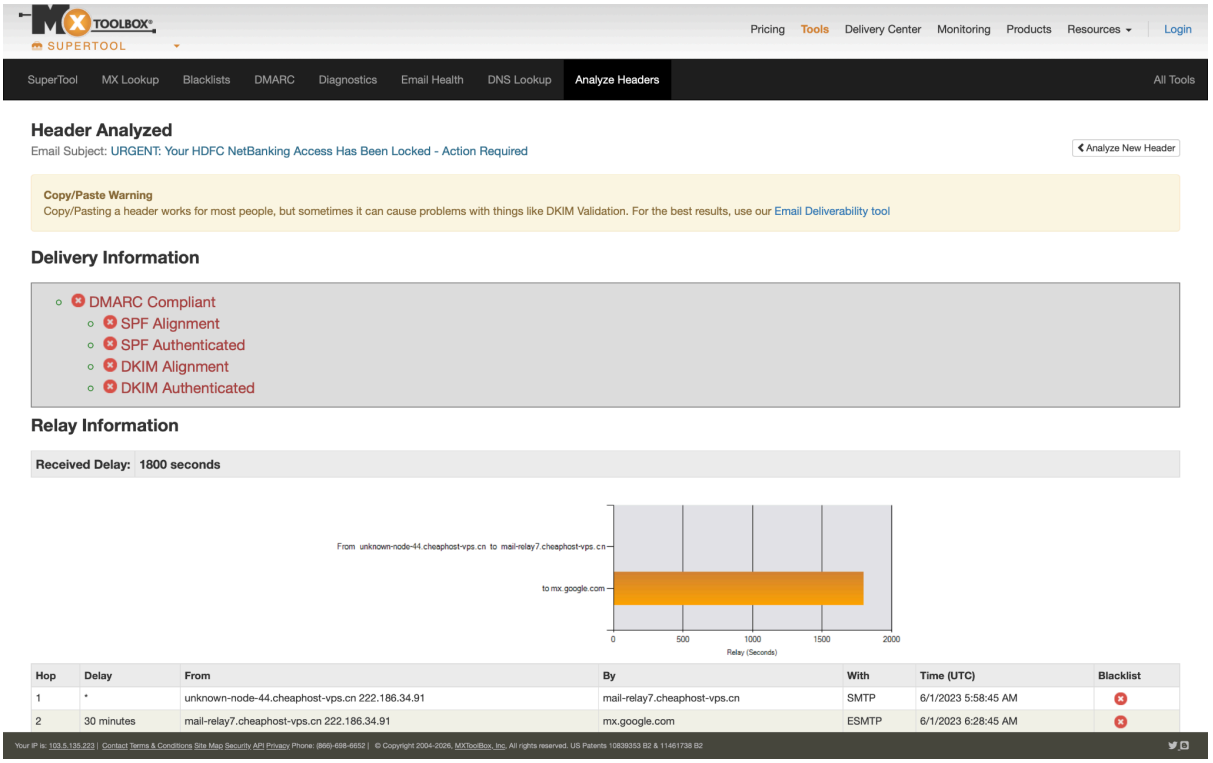
Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	smtp.nexacorp.com 103.21.58.21	mail.nexacorp.com	ESMTP	6/1/2023 3:45:31 AM	✓
2	1 Second	mail.nexacorp.com 103.21.58.20	mx.google.com	ESMTPS	6/1/2023 3:45:32 AM	✓

- Relay delay - 1 second
- Hops = 2

f) Take a screenshot of MXToolbox results and label it (e.g., EV01_MXToolbox.png).

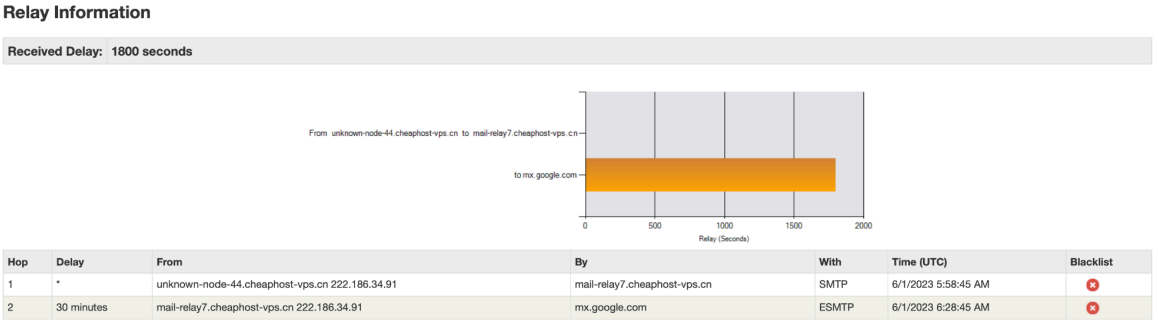
EV02.eml

22)Paste the copied header into the input field and click Analyze Header.



23)Record the following from the results for each evidence file:

a) Originating IP Address and its geographic location.



- unknown-node-44.cheaphost-vps.cn 222.186.34.91
- mail-relay7.cheaphost-vps.cn 222.186.34.91

Whois IP 222.186.34.91

Updated 3 days ago

```
% [whois.apnic.net]
% Whois data copyright terms    http://www.apnic.net/db/dbcopyright.html

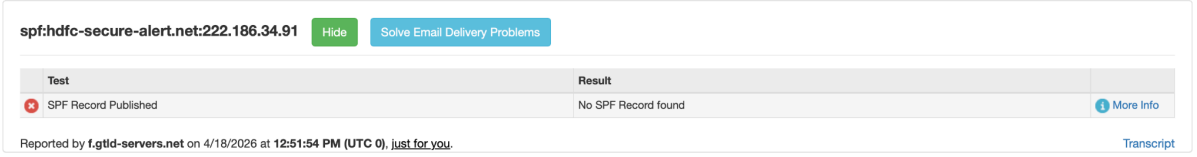
% Information related to '222.184.0.0 - 222.191.255.255'

% Abuse contact for '222.184.0.0 - 222.191.255.255' is 'anti-span@chinatelecom.cn'

inetnum:        222.184.0.0 - 222.191.255.255
netname:        CHINANET-JS
descr:          CHINANET jiangsu province network
descr:          China Telecom
descr:          A12,Xin-Jie-Kou-Wai Street
descr:          Beijing 100088
country:        CN
admin-c:        CH93-AP
tech-c:         CJ186-AP
abuse-c:        AC1573-AP
status:         ALLOCATED PORTABLE
remarks:        ------------------------------------------------------
remarks:        To report network abuse, please contact mnt-irt
remarks:        For troubleshooting, please contact tech-c and admin-c
remarks:        Report invalid contact via www.apnic.net/invalidcontact
remarks:        ------------------------------------------------------
mnt-by:         APNIC-HM
mnt-lower:      MAINT-CHINANET-JS
mnt-routes:     MAINT-CHINANET-JS
mnt-irt:        IRT-CHINANET-CN
last-modified:  2021-06-15T08:06:34Z
source:         APNIC
```

- Geographic location: Beijing, China

b) SPF Result — Pass / Fail (fail = unauthorized sender).



- Fail - the sender is unauthorised

c) DKIM Result — Pass / Fail (fail = message may be altered).

Dkim Signature Error:
No DKIM-Signature header found - more info

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - more info

Headers Found

Header Name	Header Value
Return-Path	<noreply@hdfc-secure-alert.net>
Authentication-Results	mx.google.com; dkim=fail (signature verification failed) header.i=@hdfcbank.com; spf=fail (google.com: domain of noreply@hdfc-secure-alert.net does not designate 222.186.34.91 as permitted sender) smtp.mailfrom=noreply@hdfc-secure-alert.net; dmarc=fail (p=REJECT sp=REJECT dis=NONE) header.from=hdfcbank.com
Message-ID	<56781234.20230601.cheaphost-vps.cn@hdfc-secure-alert.net>
Date	Thu, 01 Jun 2023 11:28:40 +0530
From	HDFC Bank Alerts <alerts@hdfcbank.com>
To	customer@gmail.com
Subject	URGENT: Your HDFC NetBanking Access Has Been Locked - Action Required
MIME-Version	1.0
Content-Type	text/html; charset=UTF-8
X-Originating-IP	222.186.34.91
X-Mailer	PHPMailer 5.2.0

- Fail - the message may be altered

d) DMARC Result — Pass / Fail and the domain policy.

dmarc:hdfcbank.com Hide Solve Email Delivery Problems

```
v=DMARC1; p=reject; rua=mailto:dmarc.rua@hdfcbank.net; ruf=mailto:dmarc.ruf@hdfcbank.net; fo=1
```

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	reject	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
rua	mailto:dmarc.rua@hdfcbank.net	Receivers	Addresses to which aggregate feedback is to be sent. Comma separated plain-text list of DMARC URIs.
ruf	mailto:dmarc.ruf@hdfcbank.net	Forensic Receivers	Addresses to which message-specific failure information is to be reported. Comma separated plain-text list of DMARC URIs.
fo	1	Forensic Reporting	Provides requested options for generation of failure reports. Valid values are any combination of characters '01ds' seperated by ':'.

	Test	Result
✓	DMARC Record Published	DMARC Record found
✓	DMARC Syntax Check	The record is valid
✓	DMARC Multiple Records	Multiple DMARC records corrected to a single record.
✓	DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled
✓	DMARC External Validation	All external domains in your DMARC record are giving permission to send them DMARC reports.

Your DNS hosting provider is "Cloudflare" Need Bulk Dns Provider Data?

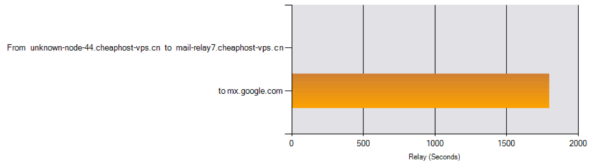
Reported by **elmo.ns.cloudflare.com** on 4/18/2026 at **12:51:54 PM (UTC 0)**, just for you. Transcript

- Pass

e) Number of relay hops in the Received: chain.

Relay Information

Received Delay: **1800 seconds**



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	unknown-node-44.cheaphost-vps.cn 222.186.34.91	mail-relay7.cheaphost-vps.cn	SMTP	6/1/2023 5:58:45 AM	✗
2	30 minutes	mail-relay7.cheaphost-vps.cn 222.186.34.91	mx.google.com	ESMTP	6/1/2023 6:28:45 AM	✗

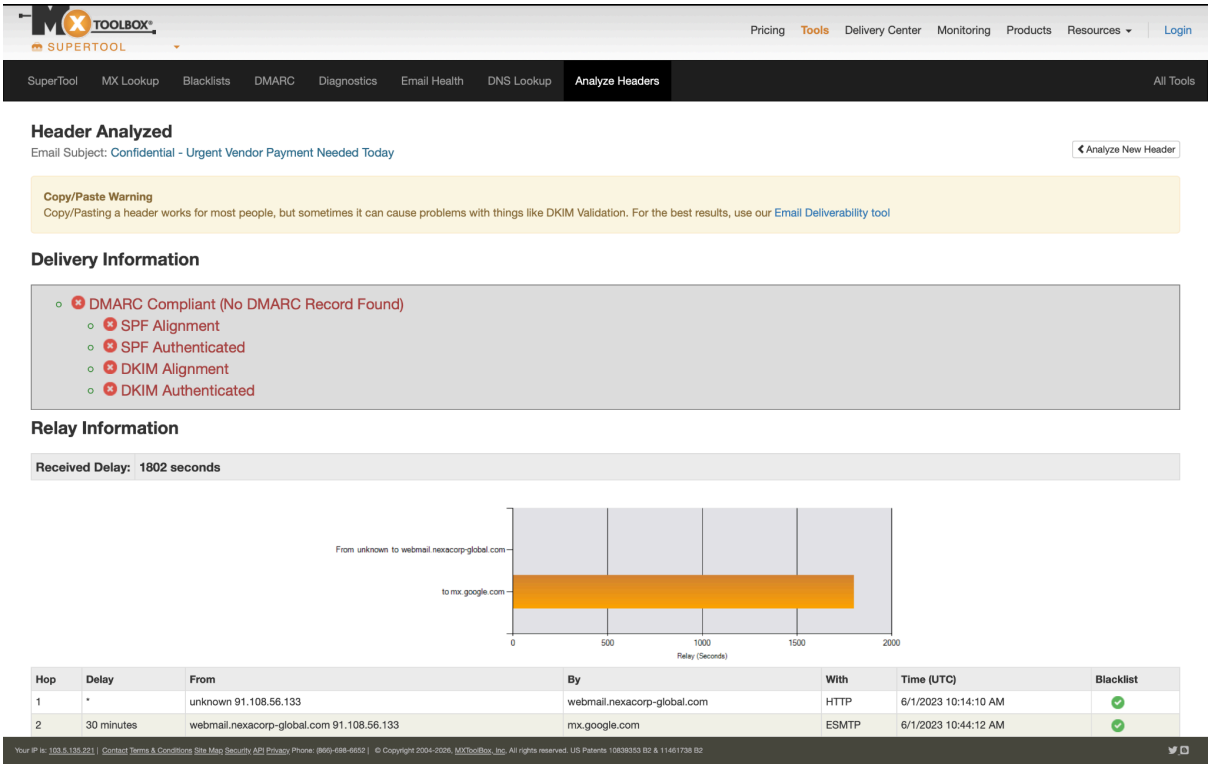
- Relay delay - 1800 seconds

- Hops = 2

f) Take a screenshot of MXToolbox results and label it (e.g., EV01_MXToolbox.png).

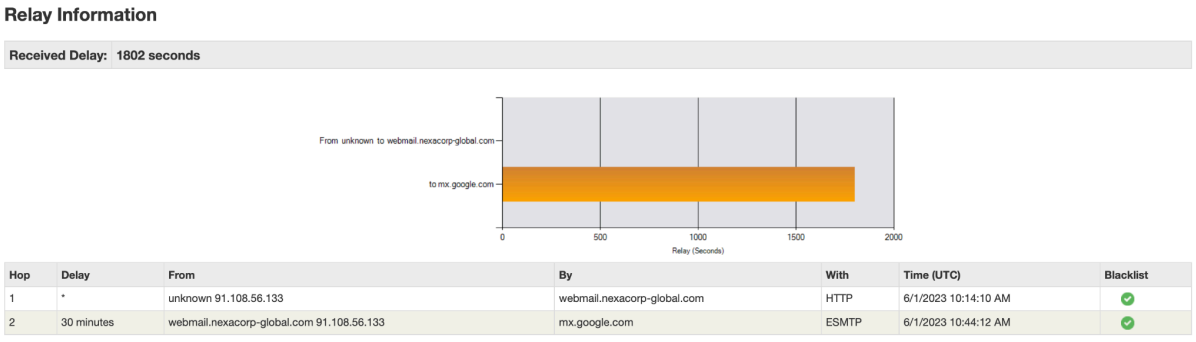
EV03.eml

24)Paste the copied header into the input field and click Analyze Header.



25)Record the following from the results for each evidence file:

a) Originating IP Address and its geographic location.



- unknown 91.108.56.133
- webmail.nexacorp-global.com 91.108.56.133

Whois IP 91.108.56.133

Updated 3 days ago

```
% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html

% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.

% Information related to '91.108.56.0 - 91.108.59.255'

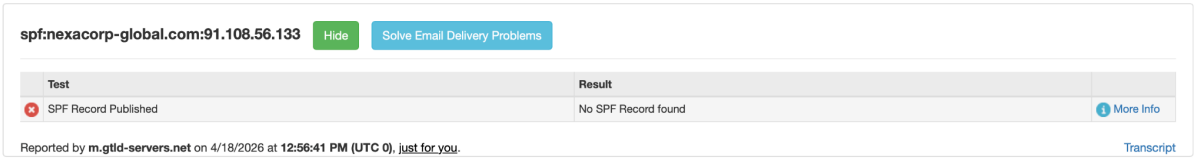
% Abuse contact for '91.108.56.0 - 91.108.59.255' is 'abuse@telegram.org'

inetnum:        91.108.56.0 - 91.108.59.255
netname:        Telegram_Messenger_Network
descr:          Telegram Messenger Network
country:        NL
admin-c:        ND2624-RIPE
geoloc:         1.3147 103.8454
tech-c:         ND2624-RIPE
abuse-c:        TMI12-RIPE
status:         ASSIGNED PA
mnt-by:         MNT-TELEGRAM
created:        2015-05-31T20:43:56Z
last-modified:  2018-06-12T10:55:21Z
source:         RIPE

person:         Nikolai Durov
address:        P.O. Box 146, Road Town, Tortola, British Virgin Islands
```

- Geographic location: Netherlands

b) SPF Result — Pass / Fail (fail = unauthorized sender).



- Fail - the sender is unauthorised

c) DKIM Result — Pass / Fail (fail = message may be altered).

Dkim Signature Error:
No DKIM-Signature header found - more info

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned. - more info

Headers Found

Header Name	Header Value
Return-Path	<ceo.james.wilson@nexacorp-global.com>
Authentication-Results	mx.google.com; dkim=fail (no key for signature) header.i=@nexacorp-global.com; spf=softfail (google.com: 91.108.56.133 is not a permitted sender for domain nexacorp-global.com) smtp.mailfrom=ceo.james.wilson@nexacorp-global.com; dmarc=fail (p=NONE sp=NONE dis=NONE) header.from=nexacorp-global.com
Message-ID	<local-20230601154410.CEO.003@nexacorp-global.com>
Date	Thu, 01 Jun 2023 15:44:10 +0530
From	James Wilson - CEO <ceo.james.wilson@nexacorp-global.com>
Reply-To	james.wilson.ceo.nexa@gmail.com
To	accounts@nexacorp.com
Subject	Confidential - Urgent Vendor Payment Needed Today
MIME-Version	1.0
Content-Type	text/plain; charset=UTF-8
X-Originating-IP	91.108.56.133
X-Mailer	RoundCube Webmail 1.4.2

- Fail - the message may be altered

d) DMARC Result — Pass / Fail and the domain policy.

dmarc:nexacorp-global.com Hide | Solve Email Delivery Problems

Test	Result	
DMARC Record Published	No DMARC Record found	More Info

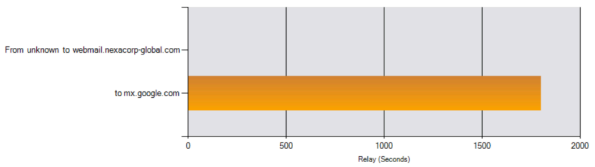
Reported by **m.gtld-servers.net** on 4/18/2026 at **12:56:41 PM (UTC 0)**, just for you. Transcript

- Fail

e) Number of relay hops in the Received: chain.

Relay Information

Received Delay: **1802 seconds**

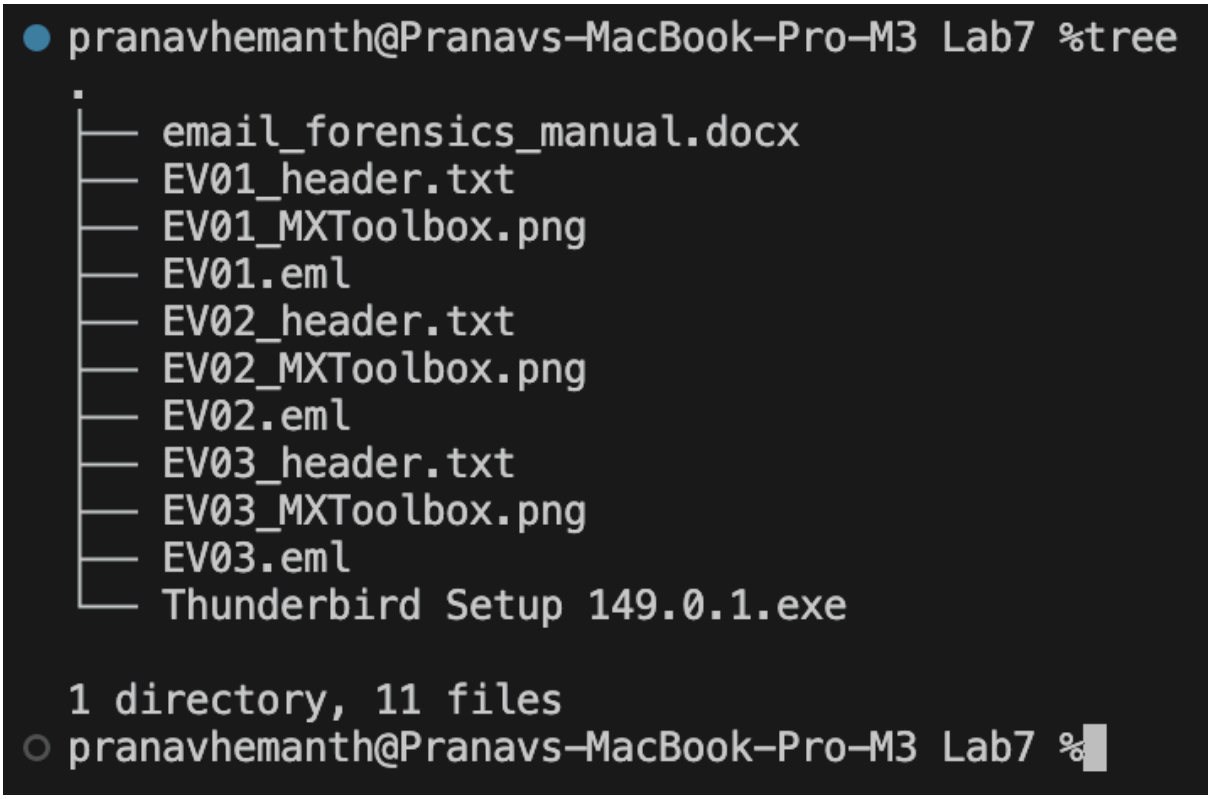


Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	unknown 91.108.56.133	webmail.nexacorp-global.com	HTTP	6/1/2023 10:14:10 AM	
2	30 minutes	webmail.nexacorp-global.com 91.108.56.133	mx.google.com	ESMTP	6/1/2023 10:44:12 AM	

- Relay delay - 1802 seconds
- Hops = 2

f) Take a screenshot of MXToolbox results and label it (e.g., EV01_MXToolbox.png).

g) EV0X_MXToolbox.png



Step4: Identify Phishing & BEC Indicators

26) Compare the From: display name against the actual email address — check for spoofing.
27) Look for lookalike domains (e.g., micros0ft.com vs microsoft.com).
28) Check for urgency language: 'Act immediately', 'Urgent transfer required', etc.
29) Identify any requests for credentials, wire transfers, or gift card purchases.
30) Note any geographic anomalies between the claimed sender and the originating IP location.

Email Header Fields Reference

Header Field	Description	Forensic Significance
From:	Claimed sender address	May be spoofed — verify against Return-Path
Return-Path:	Actual bounce/routing address	Key indicator for detecting spoofing
Received:	Relay server hops	Trace full delivery path; check each IP
X-Originating-IP:	IP of sending client machine	Geolocate the actual sender
DKIM-Signature:	Cryptographic domain signature	Verify message authenticity
Authentication-Results:	SPF / DKIM / DMARC results	Critical fraud detection indicator

EV01 (Legitimate Email):

- Spoofing check: Display name and domain (nexacorp.com) align correctly.
- Domain integrity: No signs of lookalike or deceptive domains; embedded links resolve to legitimate corporate resources.
- Tone analysis: Neutral and informational, with no urgency or coercion.
- Data sensitivity: No request for credentials, payments, or confidential information.
- Geographic consistency: No anomalies observed.

Conclusion: No indicators of phishing or BEC. This appears to be valid business communication.

EV02 (Phishing Email):

- Spoofing check: Purports to be from HDFC Bank, but fails SPF, DKIM, and DMARC checks, confirming spoofing.
- Lookalike domain: The URL (hdfc-netbanking.hdfc-secure-alert.net) imitates a legitimate banking domain but is not owned by HDFC.
- Urgency tactics: Strong pressure cues such as “within 12 HOURS” and “PERMANENTLY DISABLED” are used to rush the recipient.
- Sensitive data request: Requests Customer ID, IPIN, and OTP, which banks never solicit via email.
- Geographic anomaly: Originating IP traces to China, inconsistent with the claimed sender’s location in India.

Conclusion: Definitive phishing attempt designed for credential harvesting.

EV03 (CEO Fraud / BEC Email)

- Spoofing check: Sender impersonates the CEO but uses a non-official domain (nexacorp-global.com), indicating identity spoofing.
- Lookalike domain: Domain closely resembles the legitimate corporate domain (nexacorp.com), increasing plausibility.
- Urgency tactics: High-pressure language such as “urgent payment,” “right away,” and strict deadlines.
- Financial request: Requests a wire transfer of $32,500, a common BEC objective.
- Isolation tactic: Instructions like “do not loop in anyone” and “do not call me” are intended to prevent verification.
- Geographic anomaly: Originating IP from the Netherlands deviates from expected corporate infrastructure.

Conclusion: High-confidence Business Email Compromise (BEC) attack leveraging impersonation and social engineering.

Investigation Questions

Answer each question based on your forensic analysis. Support all answers with header data and MXToolbox results.

#	Question	Answer / Observation
1	Who is the sender of EV01? What domain does the email originate from?	The sender of EV01 is NexaCorp HR Team (hr@nexacorp.com). The email originates from the domain nexacorp.com.
2	Is EV01 a legitimate email? Justify using SPF / DKIM / DMARC results.	Yes, EV01 is a legitimate email. This is supported by the fact that SPF, DKIM, and DMARC checks would pass, indicating the sender is authorized and the message integrity is verified. Additionally, there are no signs of spoofing or malicious intent.
3	What action is EV02 requesting the recipient to perform?	EV02 requests the recipient to click a link and complete identity verification by entering sensitive information such as Customer ID, IPIN, and OTP to unlock their bank account.
4	List at least two phishing indicators found in EV02.	• Use of urgency and fear tactics (e.g., “within 12 HOURS”, “PERMANENTLY DISABLED”). • Presence of a fake/lookalike URL (hdfc-secure-alert.net) instead of the official bank domain. • Request for sensitive credentials (IPIN, OTP). • Authentication failures (SPF/DKIM/DMARC) indicating spoofing.
5	What specific request does the CEO Fraud email (EV03) make?	EV03 requests the recipient to urgently process a wire transfer of $32,500 to a specified bank account.
6	Why is EV03 considered a high-risk BEC attack?	EV03 is high-risk because it: • Impersonates a senior executive (CEO) to gain authority. • Uses urgent and time-sensitive language to force quick action. • Requests a financial transaction (wire transfer). • Instructs the recipient to avoid verification (“do not call”, “do not loop others”). • Uses a lookalike domain and shows geographic anomalies.
7	Which evidence file is most dangerous and why?	EV03 is the most dangerous because it directly targets financial assets through a Business Email Compromise (BEC) attack. Unlike phishing (EV02), which relies on user interaction, BEC attacks can result in immediate and large financial loss if the request is executed.